# PILLSYNC

## Intelligent Medicine Reminder and Medication Tracking Platform

### Milestone-4 Final Project Documentation

Project Phase: Final Implementation, Testing & Deployment

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1. Project Overview

PillSync is an intelligent medicine reminder and medication tracking platform designed to help users manage medicine schedules, monitor medication adherence, track medicine history, and receive timely refill alerts. The platform is intended to support personal healthcare management, family medication tracking, caregiver monitoring, elderly care, and chronic disease treatment management.

The system combines a web-based user interface with backend services and intelligent medication workflows. It provides a centralized platform where users can manage medicines, view medication information, monitor dosage activity, and access adherence and refill-related insights.

## 2. Problem Statement

Managing multiple medicines manually can lead to missed doses, incorrect dosage timing, difficulty maintaining medication history, and delays in purchasing medicines before the available stock runs out. Caregivers may also find it difficult to monitor medication activities for assigned patients. PillSync addresses these challenges by providing a centralized digital platform for medicine management, reminders, adherence tracking, refill monitoring, and caregiver support.

## 3. Project Objectives

- Provide a centralized platform for managing medicines and medication schedules.
- Help users receive timely medicine reminders and record taken or missed doses.
- Track medication adherence and maintain medication history.
- Monitor medicine quantity and provide refill-related alerts and predictions.
- Provide dashboards and analytics for medication and adherence information.
- Support caregiver monitoring and notification workflows.
- Provide an easy-to-use web interface for patients, caregivers, and administrators.

## 4. Proposed Solution

PillSync provides an integrated medication management workflow. A user can authenticate into the platform, manage profile information, add medicines and dosage details, view medication schedules, monitor medicine consumption, and review medication history. The platform also supports refill monitoring by using medicine quantity and dosage information to estimate when stock may be depleted and generate refill-related alerts. Caregiver and administrative workflows extend the system for monitoring and management.

## 5. Key Functionalities

**User Authentication & Role-Based Access:** Provides secure access and separates patient, caregiver, and administrative responsibilities.

**User Profile & Medication Management:** Allows users to manage profile information and maintain medicine-related details.

**Medicine Management:** Supports adding, viewing, organizing, and tracking medicines and dosage information.

**Dosage Scheduling:** Supports medication schedules and dosage-related tracking.

**Smart Reminders:** Provides scheduled medicine reminders and supports taken, missed, and snoozed reminder actions.

**Medication Adherence Tracking:** Tracks taken and missed medicines and supports adherence percentage and trend analysis.

**Refill Prediction & Alerts:** Uses medicine quantity and dosage information to estimate stock depletion and generate refill alerts.

**Medication History:** Maintains a record of medication activity for review.

**Dashboard & Analytics:** Displays medication status, adherence information, refill information, and related insights.

**Caregiver Monitoring:** Allows caregivers to monitor assigned patients and receive relevant medication notifications.

**OCR-Based Medicine Recognition:** Supports extraction of medicine information from uploaded medicine images or prescriptions as part of the planned intelligent workflow.

## 6. System Workflow

The overall PillSync workflow can be represented as follows:

**User / Caregiver/Admin**

↓

**Authentication**

↓

**Dashboard**

↓

**Medicine & Profile Management**

↓

**Dosage Scheduling**

↓

**Medicine Reminder**

↓

**Taken / Missed / Snooze**

↓

**Adherence & Medication History**

↓

**Medicine Stock Monitoring**

↓

**Refill Prediction & Alerts**

↓

**Analytics / Caregiver Monitoring**

## 7. System Architecture

The PillSync architecture is organized into presentation, application/API, data, intelligent processing, notification, and deployment layers. The frontend provides the user interface and communicates with backend APIs. The backend manages authentication, medication workflows, business logic, and database operations. Intelligent components support OCR and medication-related analysis, while notification services support reminders and alerts.

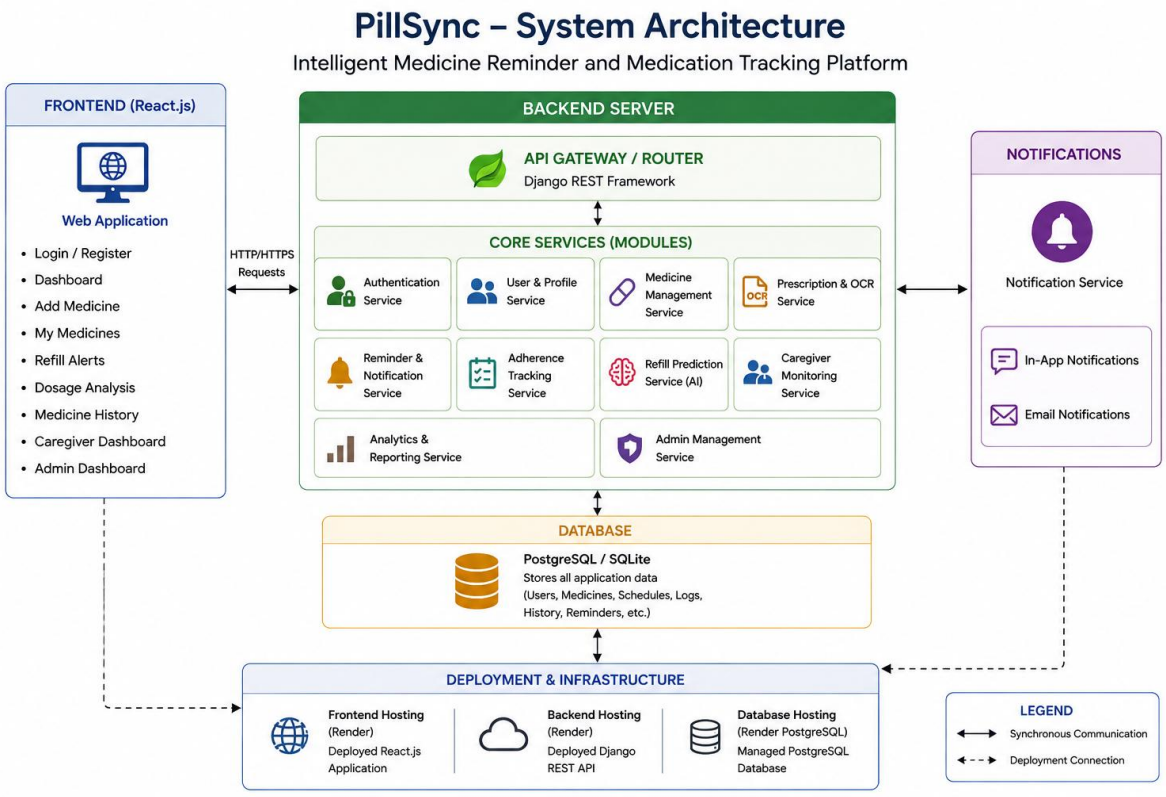



**FIG**: SYSTEM ARCHITECTURE DIAGRAM

## 8. Technology Stack

| Category | Technology | Purpose |
|---|---|---|
| Frontend | HTML, CSS, JavaScript | Web-based user interface |
| Backend | Python, Django REST Framework / FastAPI | API and application services |
| Database | PostgreSQL / SQLite (Development) | Storage of project data |
| AI & OCR | Tesseract OCR, spaCy, OpenAI API | OCR and intelligent processing |
| Authentication | JWT, OAuth2 | Authentication and access control |
| Styling / UI | CSS | Frontend styling and UI development |
| Development Tools | VS Code, Git, GitHub | Development and API testing |
| Testing | Django Test Client, React Testing Library | Testing and validation |

## 9. Frontend Implementation

The frontend provides the main interaction layer of PillSync. The interface is organized into authentication, dashboard, medication management, medication history, dosage analysis, refill monitoring, caregiver, and administrative views. The frontend communicates with backend services through APIs to display and update medication-related information.

- Authentication and login interface.
- Dashboard for accessing major medication-management functions.
- Medicine management and medicine details pages.
- Dosage analysis and medication schedule visualization.
- Refill alerts and medicine stock information.
- Medication history and tracking views.
- Caregiver and administrative dashboard interfaces.
- Navigation and reusable UI components.
- Integration of frontend pages with backend API services.

## 10. User Interface Screenshots

The following screenshots should be added from the final working application. Each screenshot should include a short caption describing the screen and its purpose.

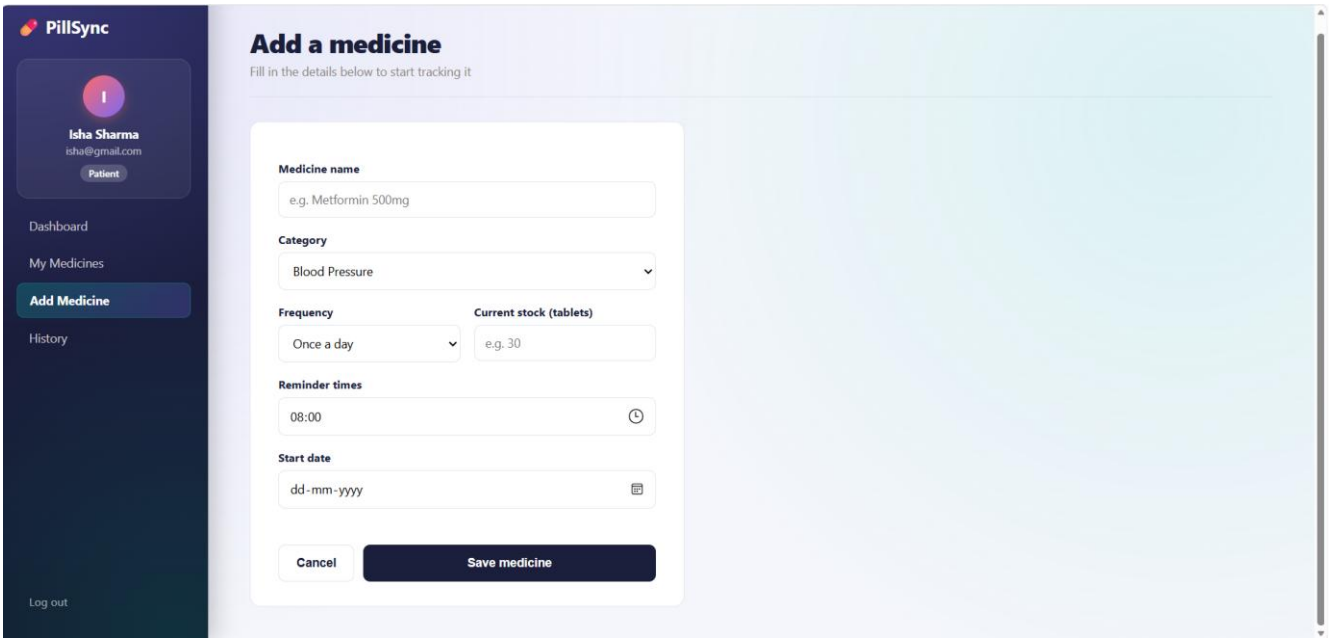
## LOGIN / AUTHENTICATION PAGE



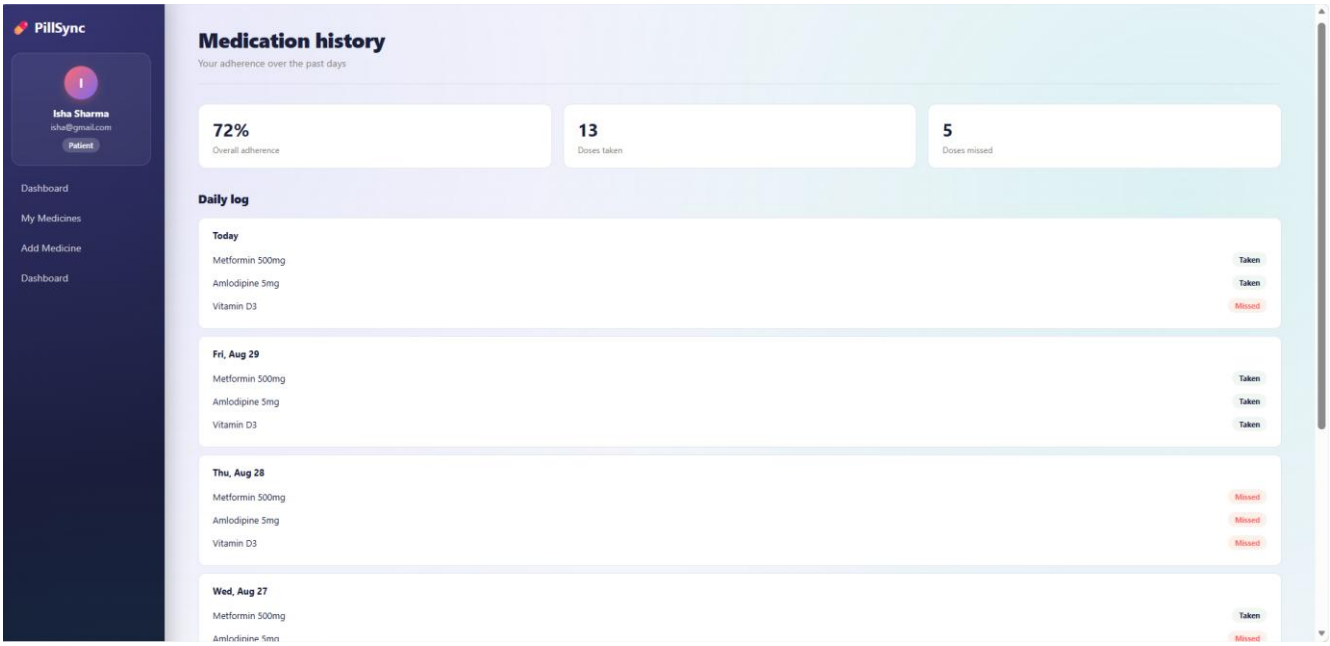

## MAIN DASHBOARD



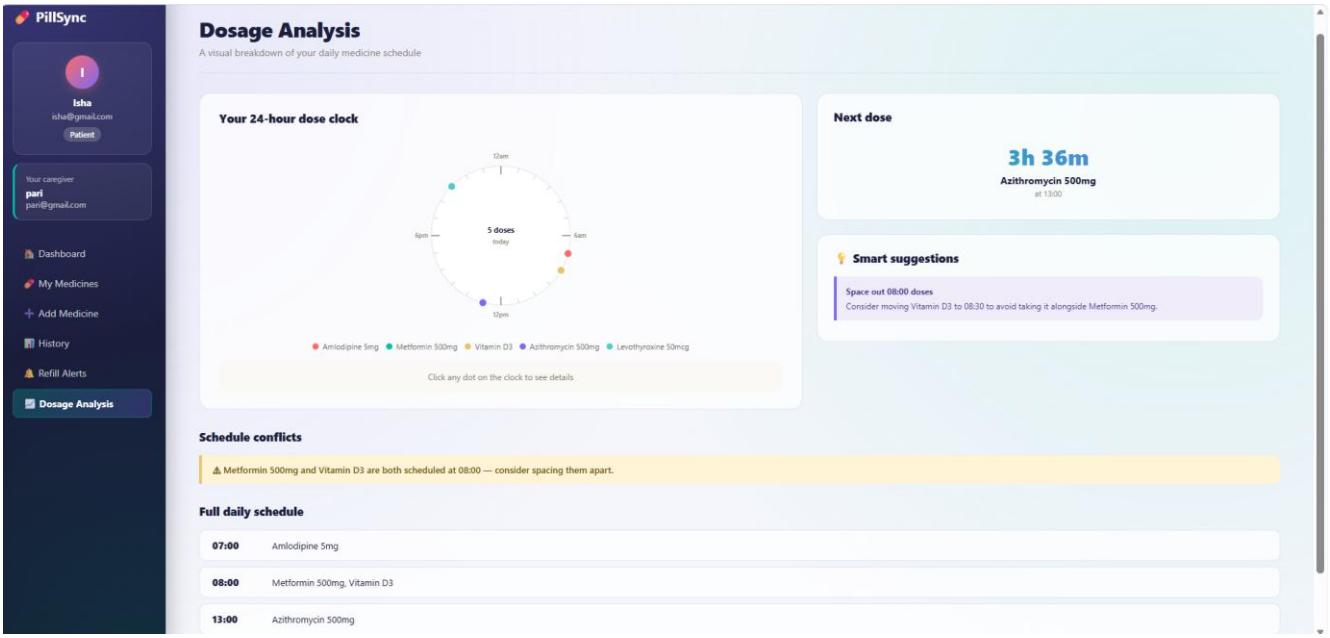
## ADD MEDICINE  PAGE



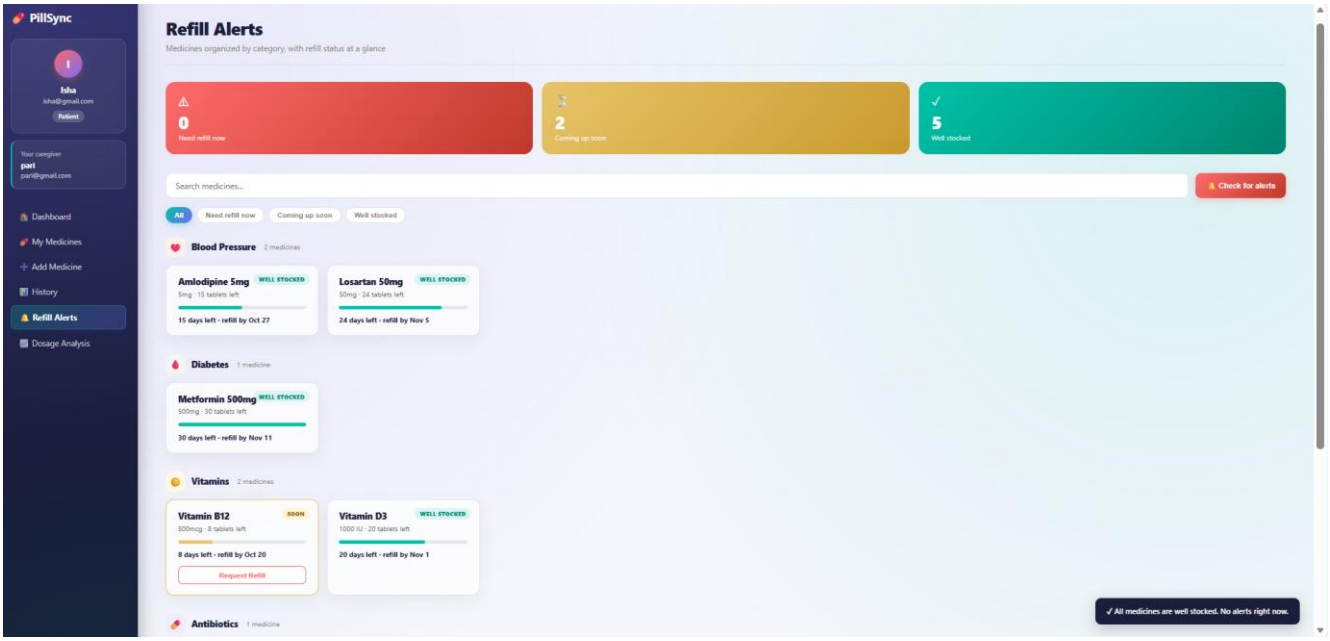

## MEDICATION HISTORY



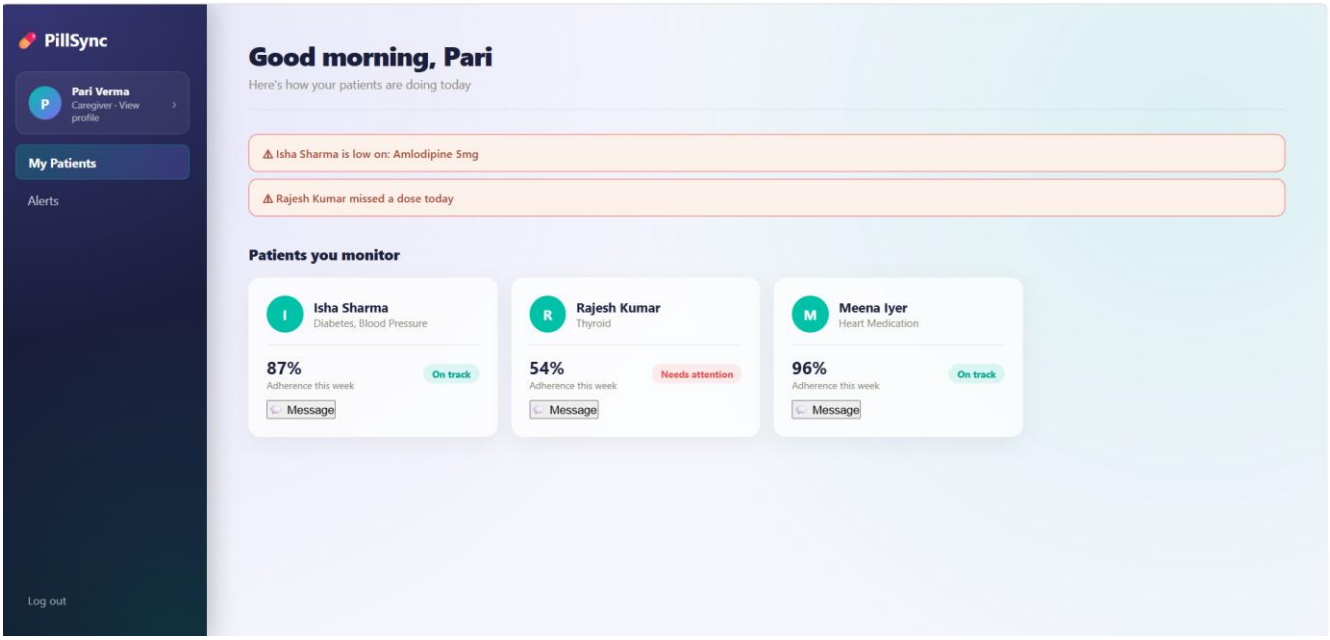
## DOSAGE ANALYSIS PAGE



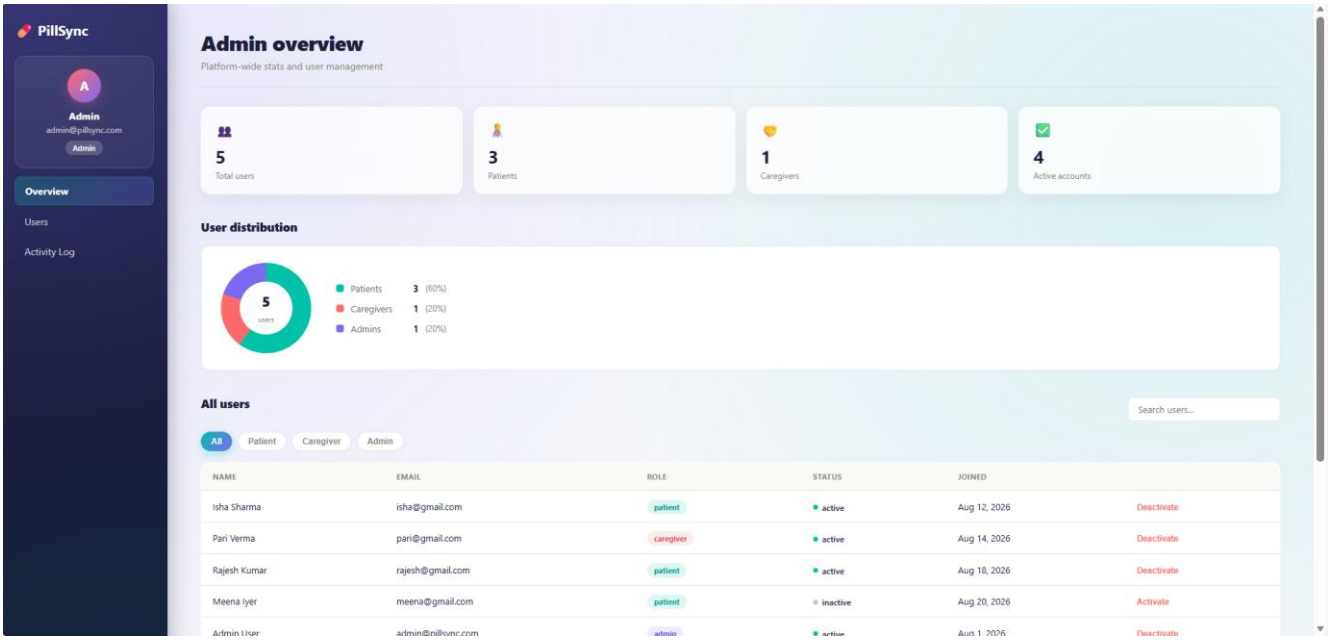

## REFILL ALERTS PAGE



**[ INSERT CAREGIVER DASHBOARD SCREENSHOT HERE ]**




**ADMIN DASHBOARD**



## 11. Frontend–Backend Integration

The frontend and backend communicate through API services. The frontend sends requests for authentication, medication information, profile information, and related workflows and receives responses that are rendered in the corresponding interface. This separation allows the UI layer and application logic to be maintained independently.

- Authentication request and user session handling.
- Fetching and displaying medicine information.
- Adding and updating medication records.
- Fetching medication history and dosage-related information.
- Displaying refill and adherence information received from backend services.
- Handling API responses and presenting relevant status information in the UI.

## 12. Testing & Validation

Testing is performed to verify that the major user workflows operate correctly and that the frontend communicates properly with backend services. The final testing phase covers authentication, medicine management, navigation, dosage-related views, refill information, history, and integrated workflows.

| Test Case | Expected Result | Actual Result | Status |
|---|---|---|---|
| User Login | Valid user can access the application | User successfully logged in | Pass |
| Dashboard Navigation | Required dashboard sections open correctly | User navigated successfully to dashboards | Pass |
| Add Medicine | Medicine details are accepted and stored | Medicine details were successfully added and displayed | Pass |
| View Medicines | Medicine list is displayed correctly | Added medicines were successfully displayed in the medicine list | Pass |
| Dosage Analysis | Dosage-related information is displayed | Dosage details displayed successflly | Pass |
| Refill Alerts | Refill/stock information is displayed | Refill related information is working successfully | Pass |
| Medication History | Medication history is accessible | Medication History is accessible and diplayed clearly | Pass |
| Backend Integration | Frontend receives and displays API data | Backend connected successfully | Pass |

## 13. Deployment

The completed application is intended to be deployed using a cloud environment so that the frontend and backend can be accessed through a live application URL. Docker and cloud deployment platforms such as AWS or Azure are included in the project technology stack.

**Live Application:** https://pillsync-med.onrender.com

**Git Repository:** https://github.com/GKSJ-Deepvision/PillSync

## 14. Challenges & Solutions

| Challenge | Solution / Approach |
|---|---|
| Integrating frontend pages with backend services | Connected UI workflows with the available API endpoints and validated returned data. |
| Maintaining consistency across multiple pages | Used common navigation, layout patterns, and reusable frontend structures. |
| Displaying medication-related information clearly | Organized medicine, dosage, history, and refill information into dedicated views. |
| Managing changes across team branches | Used Git and GitHub branches to manage and integrate frontend work. |
| Testing integrated workflows | Validated key user flows such as login, navigation, medicine management, and data display. |

## 15. Project Outcomes

- A web-based medication management interface was developed.
- Major medication-management workflows were organized into dedicated user interfaces.
- Frontend and backend integration was established for supported application workflows.
- Medication history, dosage analysis, and refill-related interfaces were incorporated.
- Caregiver and administrative views were included as part of the overall platform.
- The final project provides a structured foundation for medication reminders, adherence tracking, and refill management.

## 16. Future Enhancements

- Improve OCR accuracy across different medicine packages and prescription formats.
- Enhance AI-based refill prediction using larger historical medication datasets.
- Add advanced adherence analytics and personalized insights.
- Provide a dedicated mobile application.
- Expand notification options and emergency caregiver workflows.
- Integrate additional healthcare and pharmacy services where appropriate.

## 17. Conclusion

PillSync is designed as an intelligent medication reminder and medication tracking platform that brings medicine management, dosage scheduling, adherence monitoring, medication history, refill alerts, analytics, and caregiver support into a single system. The project combines a user-friendly web interface with backend services, database management, intelligent OCR and medication workflows, and notification capabilities.

Through the final milestone, the project focuses on integrating the developed modules, validating the major workflows, preparing the analytics and user interfaces, and deploying the application. The resulting platform provides a structured solution for improving medication organization and supporting timely medication management.

## 18. Project Links

**Live Application:** https://pillsync-med.onrender.com

**Git Repository:** https://github.com/GKSJ-Deepvision/PillSync